

Even though AI is a very important aspect for autonomic computing to work, we do not really need to simulate conscious human thoughts as such. The whole emphasis, today, is on developing computers that can be operated intuitively with minimum human involvement. This demands a system that can crunch data in a

platform-agnostic manner. And much like the human body, this system is expected to carry out its functions and adapt to its user's requirements without the need of the user to go into minute details of its functioning.

Self-management in autonomic computing

The very core of autonomic computing systems is self-management, which aims to provide freedom from tasks of system operation and maintenance, and to make available a device that works at peak performance 24 hours a day. The day-to-day working is maintained in a dynamic environment of rapidly and constantly-changing workloads, user requirements and virus attacks, and so on.

The system can also repeatedly keep an eye on its own functioning, for example, let us say, a particular component needs to be checked for upgradation. If an error is detected, the system automatically goes back to the last error-free version, while its problem-determination algorithms work towards identifying and removing the source of the error.

The IBM autonomic computing team has broken the self-management aspect further into four dimensions, namely, self healing, self optimising, self protecting and self controlling.

Self controlling. An autonomic computing system should be able to configure and reconfigure itself under diverse and volatile conditions.

The system configuration, or set-up, and the dynamic adjustments to the configuration, in order to manage dynamic environments, must occur automatically.

Self optimising. Interestingly, an autonomic computing system is never satisfied with the status quo and is forever looking for ways to optimise its working. It monitors its constituent elements and makes adjustments to the workflow to achieve predetermined system goals.

Self healing. An autonomic computing system must act like the human body in terms of healing itself. It should be able to bounce back from everyday, as well as unforeseen, problems that might cause some of its parts to fail.

It should be able to find out existing or potential problems and then seek out an alternative way of using resources or reconfiguring the system to keep functioning efficiently.

Self protecting. The virtual world faces as many threats as those faced by a physical world. Hence, an autonomic computing system should be very well-versed in the art of self protection. It should be capable of detecting, identifying and guarding itself against different types of attacks to preserve the overall system security and integrity.

The self-learning aspect

Like human beings, computers are slowly evolving into devices that learn from their own mistakes. The concept is based on the human nervous system, particularly the way our neurons act in response to stimuli and link up with other neurons to construe information. This phenomenon enables computers to digest new information while executing a job and then make changes based on varying inputs.

In the near future, a new generation of AI systems is expected to perform tasks such as speaking and listening, among others, which humans can easily do.

There is a gradual shift from engineering computing systems to one that has several characteristics of



Fig. 3: Arnold Schwarzenegger, in the Terminator franchise, which first came out in the 1980s, has an expert self-healing computer system at its core (Image courtesy: www.hollywood.com/news/movies)

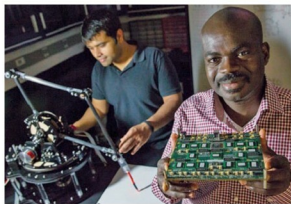


Fig. 4: A biologically-inspired processor attached to a robotic arm in a laboratory at Stanford University, the USA (Image courtesy: www.nytimes.com)

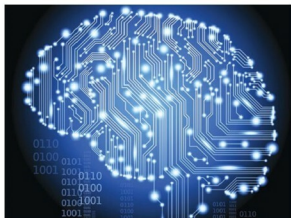


Fig. 5: A computer-simulated protein-folding image (Image courtesy: www.isgtw.org/feature/)